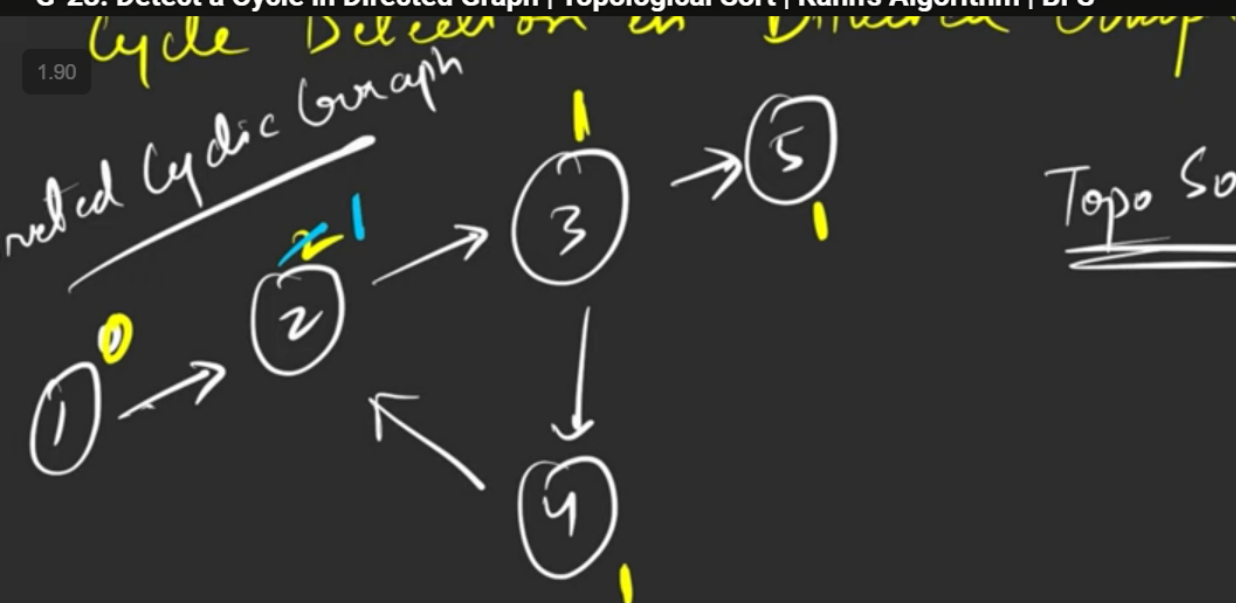




Topo Sort

$1 \rightarrow \{2\}$

$2 \rightarrow \{3\}$

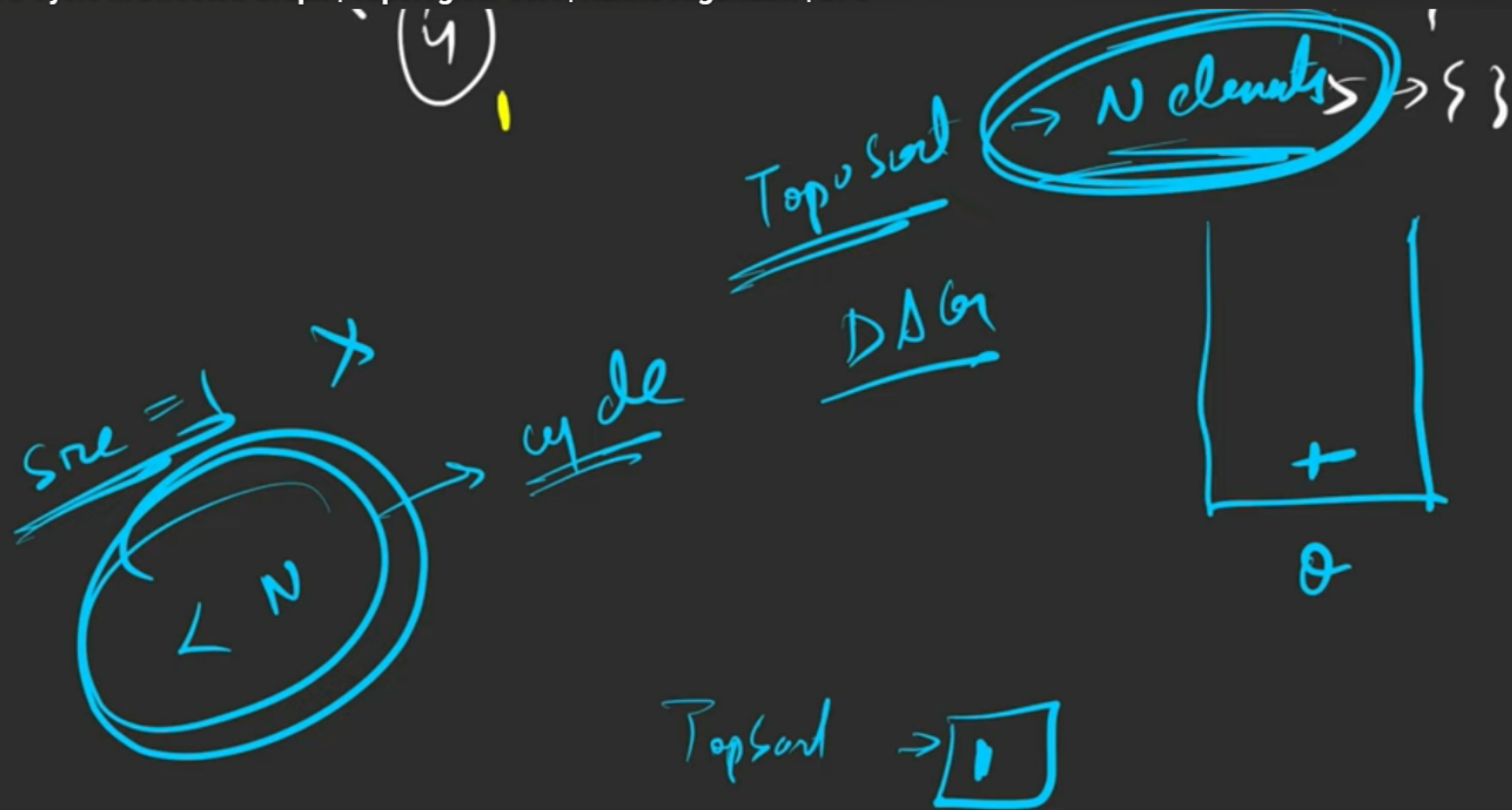
$3 \rightarrow \{4, 5\}$

$4 \rightarrow \{2\}$

$5 \rightarrow \{\}$



1.90



Topo Sort $\rightarrow$ DAG



4:31 / 6:16



G-23. Detect a Cycle in Directed Graph | Topological Sort | Kahn's Algorithm | BFS

1.90 Courses Get Hired Events </>POTD

Practice

Problem

Editorial

Submissions

Comments

C++ (g++ 5.4)

Given a Directed Graph with V vertices (Numbered from 0 to $V-1$) and E edges, check whether it contains any cycle or not.

Example 1:

Input:



Output Window

Custom Input

Compilation Results

Custom Input

Problem Solved Successfully

You get marks only for the first correct submission if you solve the problem without viewing the full solution.

Test Cases Passed:

410 / 410

Your Total Score:

359

Total Time Taken:

Correct Submission

Count:

```
6- class Solution {
7- public:
8-     // Function to detect cycle in a directed graph.
9-     bool isCyclic(int V, vector<int> adj[]) {
10-         int indegree[V] = {0};
11-         for(int i = 0; i < V; i++) {
12-             for(auto it: adj[i]) {
13-                 indegree[it]++;
14-             }
15-         }
16-
17-         queue<int> q;
18-         for(int i = 0; i < V; i++) {
19-             if(indegree[i] == 0) {
20-                 q.push(i);
21-             }
22-         }
23-
24-         int cnt = 0;
25-         // o(v + e)
26-         while(!q.empty()) {
27-             int node = q.front();
28-             q.pop();
29-             cnt++;
30-             // node is in your topo sort
31-             // so please remove it from the indegree
32-
33-             for(auto it: adj[node]) {
34-                 indegree[it]--;
35-                 if(indegree[it] == 0) q.push(it);
36-             }
37-         }
38-
39-         if(cnt == V) return false;
40-         return true;
41-     }
42- };
43- // } Driver Code Ends
```



5:30 / 6:16